BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 242 · Thinking logically: decisions and program flow

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why must a booking be saved before confirmation?

Q2. What is true AND false?

Q3. What does an if condition control?

Q4. What is false OR true?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

condition — Expression evaluated as true or false.

branch — Path selected by a decision.

AND — True only if both operands are true.

OR — True if either or both operands are true.

truth table — All Boolean input combinations and outputs.

New learning

Locate the decision — A discount needs BOTH an eligible membership and a basket of at least £50.

Follow the true branch — For member=true and basket=60, the condition is true and 10% discount produces £54.

Follow the false branch — For member=false with basket=60, the AND expression is false, so the full £60 is charged.

Worked example

A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.

1. Predict first — The age condition is true, but the ticket condition is false.

2. Model the reasoning — AND requires both true; the result is false, so the deny branch executes.

We Do · Practise with guidance

A challenge unlocks when score≥80 OR teacherOverride=true. Trace score=72 with override.

```
bool permitted = score>=80 || teacherOverride;
```

Predict first

The score condition is false because 72<80.

Model the reasoning

The override is true, so OR evaluates true and the challenge unlocks.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give A AND B for (false,false), (false,true), (true,false), (true,true).

Question 2

Access is allowed if Year13 AND (HasCode OR TeacherApproved). Evaluate Year13=true, HasCode=false, TeacherApproved=true; then Year13=false, HasCode=true, TeacherApproved=true.

Question 3

A developer changes the discount condition `member && basket>=50` to `member || basket>=50`. Give two wrong outcomes and correct the expression.

Exit Ticket · Answer before checking

Replacing AND with OR in a two-part eligibility rule can permit extra cases.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Confirmation must reflect a successful stored change.

Q2. False.

Q3. Which branch of the program runs.

Q4. True.

You Do pages · Independent questions

Q1. [4] False [1], false [1], false [1], true [1].

Q2. [6] Case 1: bracket false OR true=true [1], true AND true=true [1], allowed [1]. Case 2: bracket true OR true=true [1], false AND true=false [1], denied [1].

Q3. [6] A member spending under £50 incorrectly gets discount [2]; nonmember spending ≥£50 incorrectly gets discount [2]; correct member && basket>=50m [2].

Exit Ticket · Model answer

True. OR is true when either condition holds.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: